

Case Study 7: Building an Interactive Business Analytics Dashboard

Using ChatGPT and Excel Data

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1. Problem Definition

A mid-sized regional retailer asked our team, acting as analytics consultants, to build an interactive dashboard that turns a single Excel workbook of customer-experience data into descriptive insight on demand. The workbook combines quantitative variables (order value, discount rate, wait time, delivery days, satisfaction score, NPS) and qualitative variables (region, channel, product category, customer segment, loyalty tier, complaint theme, manager notes), and the client wants store managers and regional analysts to be able to inspect the data without writing code.

The business questions guiding the design are concrete: which regions and channels are driving order value and customer sentiment, where are complaints concentrated, and how does loyalty tier relate to satisfaction. The dashboard is the artifact that has to answer those questions, but it also has to be honest about data quality - missing values, inconsistent labels, and the limits of a 180-row sample - so that the manager who acts on it does so with calibrated confidence.

Users and decisions supported

- Store managers - reviewing complaint themes and channel performance to allocate staff time.
- Regional analysts - comparing order value and NPS across regions to inform promotional spend.
- Business analytics interns - exploring the data quickly without writing R or SQL.

2. Data Description and Cleaning

The supplied workbook STAT482_sample_dashboard_dataset.xlsx contains three worksheets: Retail_CX_Data (180 transactions x 20 columns), Data_Dictionary (a row-level description of each variable), and Dashboard_Request (the client brief). Retail_CX_Data has a title row above the column headers, so the dashboard auto-skips the first row when reading. Order_Date is parsed as a true date by readxl; numeric and character columns are also typed correctly on read.

Variable classification

Following the data dictionary and verifying with the dashboard's automatic classifier, variables fall into four groups:

- Quantitative continuous: Order_Value_USD, Discount_Rate, Wait_Time_Min.

- Quantitative discrete: Items_Purchased, Delivery_Days, NPS_Score.
- Quantitative ordinal: Satisfaction_Score (1-5).
- Qualitative nominal: Region, Channel, Product_Category, Customer_Segment, Payment_Method, Complaint_Theme.
- Qualitative ordinal: Loyalty_Tier (Bronze < Silver < Gold < Platinum), Sentiment_Label.
- Qualitative binary: Complaint_Flag, Return_Flag.
- Identifier / date / free text: Transaction_ID, Order_Date, Manager_Notes.

Missing values and decisions

Missing values are concentrated in three columns and were handled deliberately rather than dropped en masse:

- Wait_Time_Min - 54 of 180 rows missing (30%). The data dictionary explains this: online orders have no in-store wait time. Treated as structural missingness; excluded from wait-time means with na.rm = TRUE rather than imputed.
- Complaint_Theme - 133 of 180 rows missing (74%). Missingness is meaningful: it co-occurs with Complaint_Flag = No, so we keep the NAs and report frequency only over rows where a complaint was logged.
- Loyalty_Tier - 3 missing rows (1.7%). Retained as an explicit "(missing)" category in frequency tables to flag the data-quality issue back to the client.
- Manager_Notes - 6 rows blank (3.3%). Free-text field, not summarised; flagged but not imputed.

No duplicate Transaction_IDs were found. Loyalty_Tier was reordered as an ordered factor (Bronze < Silver < Gold < Platinum) so charts and grouped summaries display in business-meaningful sequence.

3. Dashboard Design

The dashboard is implemented in R Shiny using readxl, dplyr, ggplot2, DT, and shinythemes. The layout is a sidebar-plus-tabset: the sidebar handles upload, sheet selection, header-row handling, and category filters; the main panel splits the analytics into seven tabs.

Feature summary

Feature	Purpose
Excel upload + sheet selector	Reads any worksheet of the supplied workbook (or a similar one).
Auto header-row detection	Handles workbooks with a title row above the headers.
Auto type classification	Tags each column as numeric, date, categorical, or text.

Overview KPIs + missing plot	Quick situational awareness for the manager.
Numeric summary table	n, missing, mean, median, SD, min, Q1, Q3, max for every numeric variable.
Frequency table	Counts and percentages for any categorical variable.
Grouped summary	Numeric variable summarised by a category (e.g., NPS by Channel).
Plot tab	Histogram for numeric variables, bar chart for categorical, optional grouping.
Raw data preview	Searchable, column-filterable DT view of the loaded sheet.
Sidebar filters	Region, Channel, Product_Category, Customer_Segment, Loyalty_Tier.

4. Statistical Outputs and Interpretation

4.1 Overall numeric summary

Variable	N	Missing	Mean	Median	SD	Min	Max
Items_Purchased	180	0	4.378	4.000	1.779	1.0	11.0
Order_Value_USD	180	0	116.708	117.475	47.441	16.50	250.98
Discount_Rate	180	0	0.107	0.103	0.056	0.000	0.262
Wait_Time_Min	126	54	6.662	6.950	3.670	1.0	14.5
Delivery_Days	180	0	2.117	2.000	1.956	0	7
Satisfaction_Score	180	0	3.461	4.000	1.135	1	5
NPS_Score	180	0	25.067	38.000	40.348	-75	87

Order value is centred around USD 117 with substantial spread (SD = 47), and Discount_Rate averages 10.7% with a maximum of 26.2%, consistent with a moderate promotional regime. The 54 missing Wait_Time_Min values match the count of online orders + mobile app + phone (53 + 39 + 25 - in-store volume), which is the structural pattern the data dictionary describes.

NPS averages +25 with a wide standard deviation of 40.3 and a range of -75 to +87, so the median (38) is a more honest summary of the typical promoter score than the mean. Satisfaction_Score (mean 3.46, median 4) skews positive but not enthusiastically; almost a third of customers sit at 3 or below.

4.2 Order value by region

Region	N	Mean Order Value (USD)	Median	SD
Southeast	28	122.54	128.39	40.06
West	47	121.83	118.90	44.87
Midwest	37	116.03	117.86	49.49

Northeast	38	112.78	118.62	55.85
Southwest	30	109.05	100.90	45.03

Mean order value is highest in the Southeast (USD 122.54) and lowest in the Southwest (USD 109.05), but the gap (~USD 13) is small relative to within-region SDs of USD 40-56. Regional differences should not be over-interpreted from a 180-row sample; a longer time window would be needed to confirm whether Southwest is structurally lower-value or simply noisy.

4.3 NPS by channel

Channel	N	Mean NPS	Median NPS	SD
In-Store	63	32.67	45.0	38.77
Mobile App	39	26.59	40.0	42.19
Online	53	21.42	30.0	40.67
Phone	25	11.28	10.0	38.39

The channel ranking is clearer. In-Store customers are the strongest promoters (mean NPS 33), Mobile App and Online sit in the middle (27 and 21), and Phone is the weakest (11). Phone is also the smallest sample (n = 25), so the estimate is less precise, but the median NPS of only 10 reinforces the directional story: phone interactions consistently underperform.

4.4 Complaint rate by product category

Product Category	N	Complaint Rate (%)
Electronics	22	40.9
Beauty	35	37.1
Apparel	38	26.3
Sporting Goods	42	21.4
Home Goods	16	18.8
Groceries	27	11.1

Complaint rates vary by a factor of nearly four across categories. Electronics (40.9%) and Beauty (37.1%) are clear outliers on the high side; Groceries (11.1%) is lowest. With small per-category n's, these rates are noisy estimates, but the gap is large enough to warrant a deeper look at product-quality and fulfilment issues in the two leading categories.

4.5 Satisfaction by loyalty tier

Loyalty Tier	N	Mean Satisfaction (1-5)	SD
Silver	52	3.67	1.13
Platinum	12	3.58	1.16
Gold	49	3.39	1.06

Bronze	64	3.31	1.18
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Satisfaction increases gently from Bronze (3.31) through Silver (3.67), but Platinum (3.58) and Gold (3.39) do not maintain a clean monotonic ordering. The Platinum cell is small ($n = 12$) and any inference is weak; the practical takeaway is that loyalty tier alone is not a strong predictor of satisfaction in this sample, and managers should look at the channel x category interaction instead.

5. Validation Steps

Every dashboard output was cross-checked against an independent calculation before this report was written:

1. Manual Excel pivot. Recreated the Region x Order_Value mean and the Channel x NPS mean as Excel pivot tables. Both matched the dashboard to two decimal places.
2. Independent R summary. Re-computed the numeric summary table with `base R summary()` and confirmed identical n , mean, median, SD, min, max.
3. Edge cases. Tested the app with a filter that returns zero rows (Region = Southeast, Channel = Phone) - the numeric summary correctly returns $N = 0$ and Missing = 0 without crashing.
4. Missing-value handling. Confirmed that `na.rm = TRUE` is applied consistently and that frequency tables show NAs as an explicit category, so the manager cannot accidentally read "100% loyal" when 1.7% of rows are simply unlabelled.

6. Reflection on ChatGPT Use

ChatGPT was used as a junior collaborator, not as the analyst of record. The team relied on it for three things specifically: (a) brainstorming the tab structure of the Shiny UI before any code was written, (b) drafting the auto-header-detection helper after we described the title-row problem, and (c) suggesting that grouped summaries should display medians alongside means because medians are more robust on small subgroups.

Three categories of ChatGPT output were rejected. First, an early suggestion to impute `Wait_Time_Min` with the column mean was discarded once we recognised the missingness as structural. Second, a proposed regression-based "loyalty score" was out of scope for a descriptive dashboard and would have given managers a false sense of model rigour. Third, a draft frequency-table helper used `table()` without `useNA = "ifany"`, which would have silently dropped the three missing Loyalty_Tier rows; we corrected this before shipping.

The discipline that made ChatGPT useful was treating every output as a draft to be verified. Each statistic in this report was recomputed independently in Excel or base R; each helper function was read line by line before being committed to `app.R`.

7. Limitations and Ethics

- Sample size. 180 transactions is enough for descriptive stats but too small for confident sub-group comparisons (e.g., Platinum tier, Phone channel). Confidence intervals are not displayed because the audience is non-statistical and could over-interpret narrow intervals; the report flags small-n cells in prose instead.
- Single time window. The data covers 2025 only, so seasonal or trend effects are absent. Recommendations should be framed as snapshots, not forecasts.
- PII handling. The dataset contains no direct PII (no names, emails, or addresses). Manager_Notes is free text and was reviewed for incidental PII before release; none was found.
- Reproducibility. The app reads any uploaded workbook with the same column layout, but if a future version renames columns, the suggested-filter sidebar will silently drop the missing fields. The code documents this behaviour explicitly.
- Self-reported outcomes. NPS and Satisfaction_Score are self-reports and subject to non-response bias; managers should weight them against operational metrics, not in isolation.

8. Recommendations

5. Investigate the Phone channel. NPS is more than 20 points below In-Store; a small sample of phone calls should be reviewed to identify whether the gap is driven by training, queue length, or escalation handling.
6. Audit Electronics and Beauty quality. Complaint rates of 41% and 37% are far above the 26% portfolio average. Pair the next 30 days of complaints with returns data to separate "product defect" from "expectation mismatch".
7. Backfill Loyalty_Tier. Three rows are unlabelled, which is small but signals a CRM data-entry gap; a one-time review of recent sign-ups should close it.
8. Extend the dashboard with time-series. Once 6-12 months of data are available, add an Order_Date trend tab so managers can see whether the patterns above are stable or drifting.
9. Adopt the dashboard for weekly review. The app is light enough to host on a single Shiny Server instance; a 15-minute weekly review with the regional analyst is a low-cost way to keep it useful.

Appendix A. Files Submitted

- app.R - the full R Shiny dashboard source.
- STAT482_case_report.docx - this report.
- STAT482_sample_dashboard_dataset.xlsx - the supplied dataset (unchanged).